

Radical Halogenation

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1 Foundations

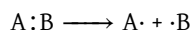
Core Concept 1.1: Radical Halogenation

Radical halogenation is a substitution reaction in which a hydrogen atom of an alkane is replaced by a halogen atom, most commonly chlorine or bromine. Unlike conventional organic reactions that proceed through ionic intermediates, radical halogenation is mediated by **free radicals**—transient, highly reactive species characterized by the presence of an unpaired valence electron.

In synthetic organic chemistry, selective atomic substitution within molecular frameworks is a fundamental objective. However, alkanes present a significant challenge due to their chemical **inertness**; comprising exclusively strong, nonpolar C–C and C–H bonds, alkanes lack reactive functional groups that serve as recognition sites for electrophilic or nucleophilic reagents. Radical halogenation addresses this limitation by converting chemically inert alkanes into functionalized **haloalkanes** (*alkyl halides*), which possess reactive carbon-halogen bonds amenable to subsequent transformations. Consequently, radical halogenation represents a critical entry point into the broader landscape of functional group interconversions.

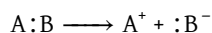
Core Concept 1.2: Homolytic vs. Heterolytic Cleavage

To understand radical formation, we must first examine bond cleavage mechanisms. Consider the **homolytic cleavage** of a generic covalent bond in compound AB.



Homolytic cleavage results in symmetric bond fission, wherein each fragment retains one electron from the original bonding pair, yielding two **radical** species. The enthalpy change (ΔH) for this process corresponds to the bond dissociation energy, typically expressed in units of $kcal\ mol^{-1}$ and denoted as DH° .

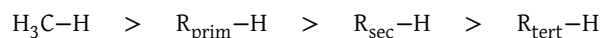
This mechanism is fundamentally distinct from **heterolytic cleavage**, in which the bonding electron pair is retained entirely by one fragment, generating a cation-anion pair:



Importantly, the energy requirement for homolytic cleavage (DH°) typically exceeds that of heterolytic processes in polar solvents, as the latter benefit from solvation stabilization of the resulting ionic species.

Core Concept 1.3: Hydrocarbon C–H Bond Types

Alkane functionalization requires **C–H bond cleavage**, and the bond dissociation energy (DH°) exhibits a systematic decrease across the following homologous series.



The designations R_{prim} , R_{sec} , and R_{tert} refer to primary, secondary, and tertiary carbon centers, which are bonded to one, two, and three additional carbon atoms, respectively. These structural classifications are illustrated in Figure 1.

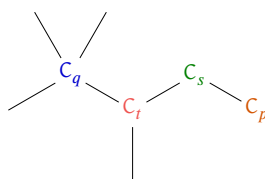


Figure 1: Primary (orange), secondary (green), tertiary (red), and quaternary (blue) carbons

The tertiary $R_{\text{tert}}-H$ bond exhibits the lowest bond dissociation energy and consequently generates the most thermodynamically stable radical upon cleavage. This stability trend—wherein radical stability increases with increasing alkyl substitution—arises from **hyperconjugation** (Core Concept 1.4).

Core Concept 1.4: Hyperconjugation

Hyperconjugation is a stereoelectronic stabilization mechanism wherein e^- density from a filled σ orbital (typically a $C-H$ or $C-C$ bond) delocalizes into an adjacent unfilled or partially filled p-orbital. In alkyl radicals, the bonding e^- s of σ bonds on substituent carbon atoms interact with the singly occupied p-orbital (SOMO) of the radical center, resulting in partial e^- delocalization that stabilizes the radical species.

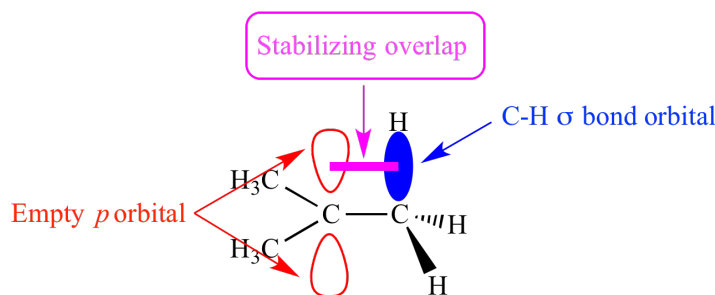


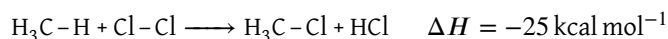
Figure 2: Hyperconjugation in alkane radicals

As illustrated in Figure 2, the magnitude of hyperconjugative stabilization is maximized in tertiary radicals ($\cdot R_{\text{tert}}$) due to the presence of a greater number of adjacent $C-H$ or $C-C$ σ bonds available for orbital overlap with the radical p-orbital. Each additional alkyl substituent provides additional filled σ orbitals capable of donating e^- density, thereby enhancing radical stability through cumulative hyperconjugative interactions.

2 Mechanism of Radical Halogenation

Core Concept 2.1: Initiation & Propagation

Having established the fundamental principles of alkane radical formation (Section 1), we now examine the mechanistic details of radical halogenation. The overall transformation, illustrated here using chlorine as the halogen, proceeds as follows.



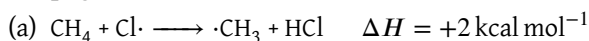
Although the reaction is net exothermic ($\Delta H < 0$), it requires an initial activation energy to proceed. To elucidate the reaction pathway, we can decompose the overall transformation into the following elementary steps.

(1) Initiation



Homolytic cleavage of the $Cl-Cl$ bond can be induced through two primary mechanisms. **Thermal energy** elevates molecular vibrational energy to levels sufficient to overcome the bond dissociation energy, resulting in bond scission. Alternatively, **photochemical activation** via UV radiation promotes a bonding e^- to an antibonding molecular orbital ($\sigma \rightarrow \sigma^*$ transition), weakening the $Cl-Cl$ bond and facilitating homolytic cleavage.

(2) Propagation



The frontier molecular orbital interactions governing this hydrogen abstraction step are depicted below in Figure 3.

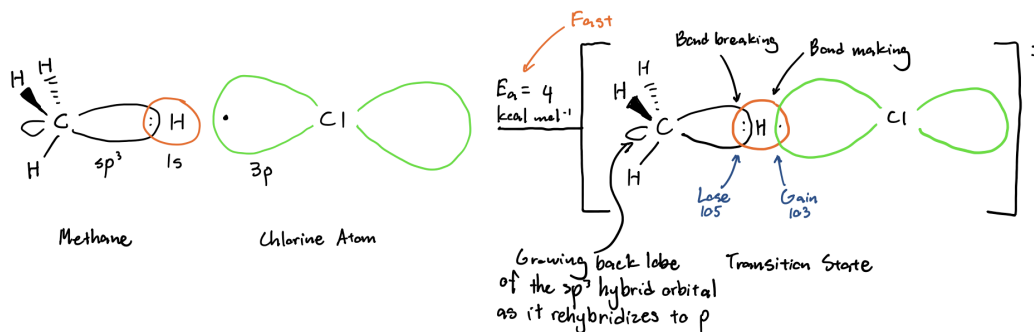


Figure 3: Orbital interaction of reagents in (2a)

This step yields a methyl radical ($\text{H}_3\text{C}\cdot$), characterized by a singly occupied 2p orbital and three sp^2 -hybridized C-H bonds adopting trigonal planar geometry, along with HCl. In aqueous or polar environments, HCl, being a strong Brønsted acid, dissociates completely into solvated H^+ and Cl^- ions.

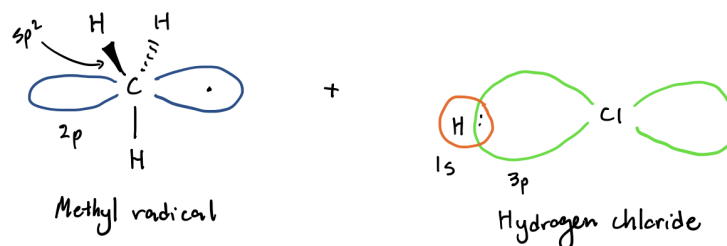
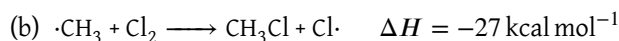


Figure 4: Products of (2a)



Summing the two propagation steps (2a) and (2b) regenerates the original stoichiometric equation.



Notably, the initiation step does not contribute to the net stoichiometry; it functions **solely to generate the initial radical species required to initiate the propagation cycle**. The chain-propagating nature of this mechanism means that a small initial population of $\text{Cl}\cdot$ radicals can facilitate the conversion of substantial quantities of starting materials, as each propagation cycle regenerates the chlorine radical. This autocatalytic regeneration enables each initiating radical to participate in numerous propagation cycles before termination, analogous to a catalytic turnover process.

The PE diagram for the propagation sequence (Figure 5) provides a quantitative representation of the energetic profile throughout the reaction coordinate.

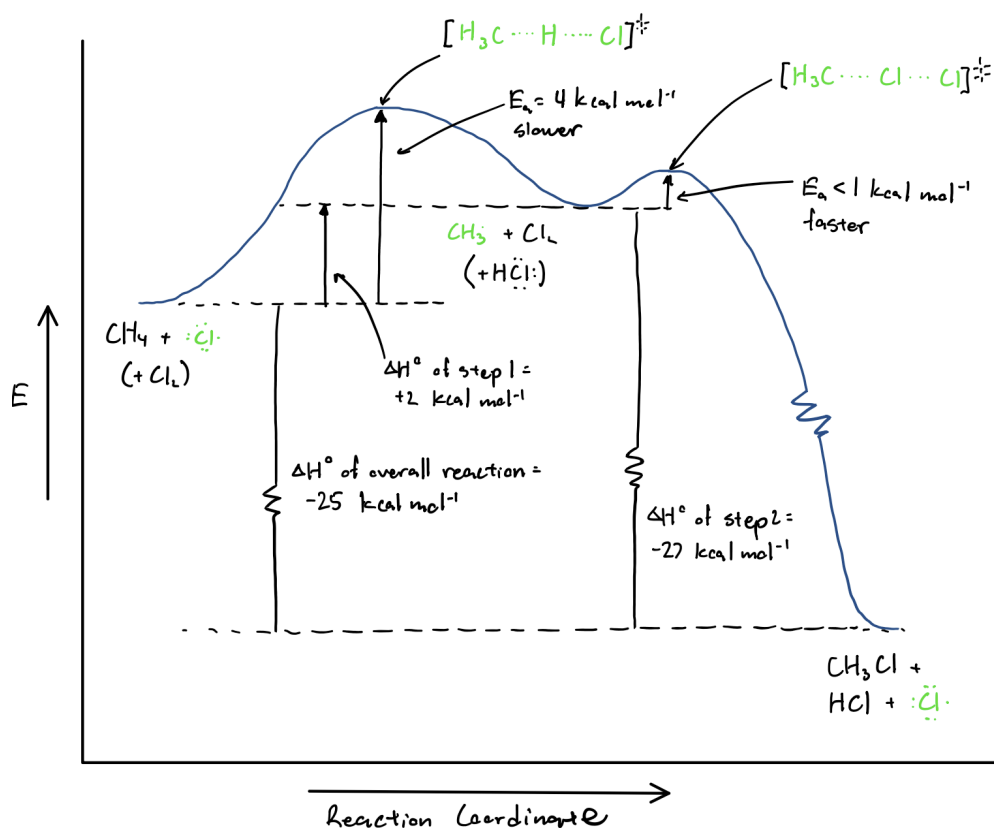
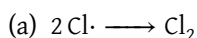


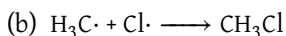
Figure 5: Potential energy diagram of halogenation propagation

Core Concept 2.2: Termination

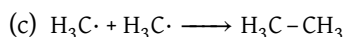
Chain termination occurs through the following radical-radical coupling reactions.



This reaction represents the microscopic reverse of the initiation step, regenerating molecular chlorine through recombination of two chlorine radicals.



While this reaction formally produces the desired chloromethane product, it terminates the chain by consuming a propagating chlorine radical without regeneration, thereby halting further catalytic turnover.



This dimerization reaction couples two methyl radicals to yield ethane, depleting the methyl radical pool required for propagation.

Throughout the reaction, radical intermediates such as $\text{Cl}\cdot$ and $\text{H}_3\text{C}\cdot$ exist at extremely low steady-state concentrations, as these transient species react rapidly with abundant substrate molecules (CH_4 and Cl_2 , respectively) rather than persisting in solution. Consequently, termination reactions—which require bimolecular encounters between two radical species—exhibit negligible rates during the early and intermediate stages of the reaction due to the low probability of simultaneous radical-radical interactions. Termination becomes significant only in the late stages of the reaction when substrate depletion reduces propagation rates, or when radical concentrations increase sufficiently within the closed system.

3 Halogen Reactivity and Selectivity

Core Concept 3.1: Reactivity of Other Halogens

Just like we have done for Cl, let's take a look at the reactivity of the other halogens (F, Br, and I).

DH° (all +)				ΔH of H – X Formation (all -)				ΔH of $H_3C - X$ Formation (all -)			
F_2	Cl_2	Br_2	I_2	HF	HCl	HBr	HI	CH_3F	CH_3Cl	CH_3Br	CH_3I
38	58	46	36	136	103	87	71	110	85	70	51

LEFT COLUMN: As you can see, the initiation dissociation energy requirements are OK for all. Seeing that the DH° for Cl_2 is the highest and we know that the reaction proceeds, we can assume that initiation would not be a barrier for any of the other halogens.

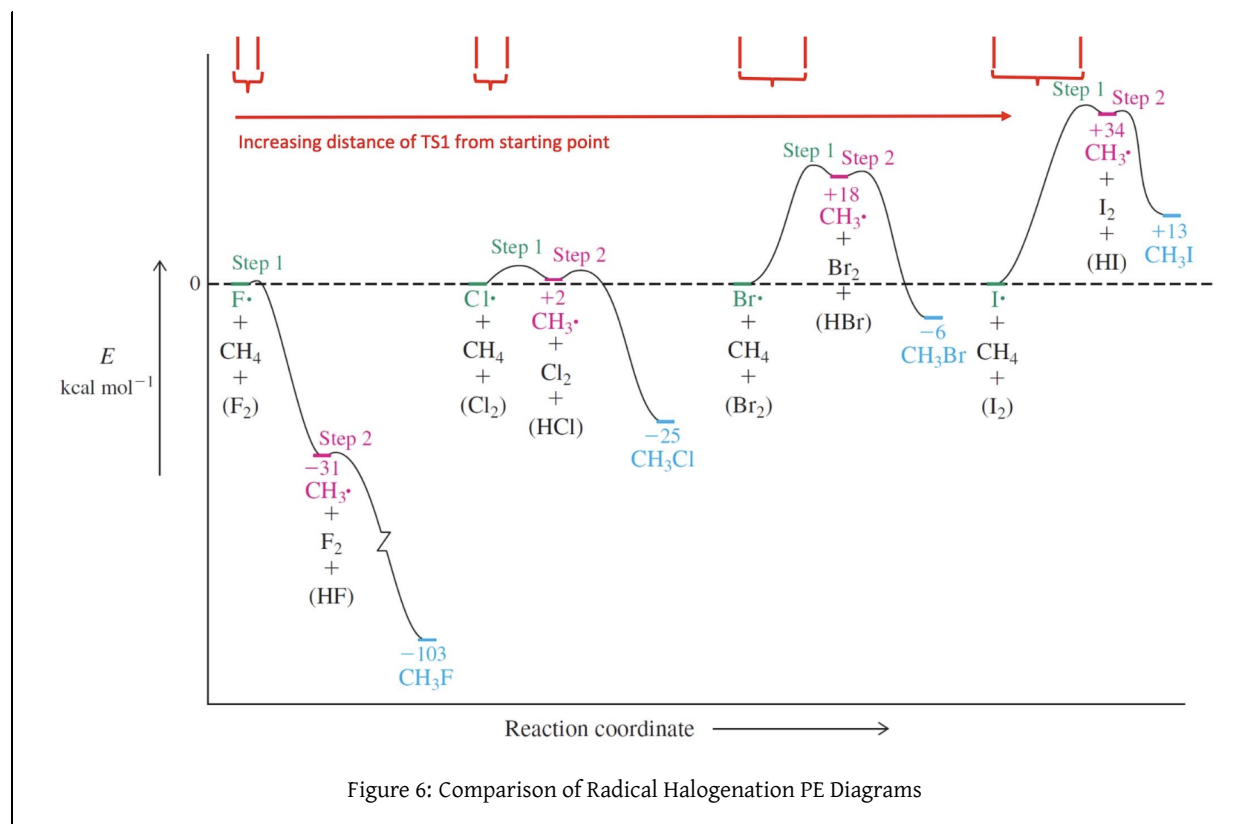
MIDDLE COLUMN: We see that the energy released from the formation of the corresponding acid for each halogen decreases as we go down the periodic table. Thus HF formation would be the most favorable and HI the least.

RIGHT COLUMN: Again, the formation of the desired product is the most favorable for F and the least favorable for I.

Based on these numbers, we can predict that radical halogenation with F will be the most reactive, followed by Cl, then Br, and finally I.

In reality, F is actually too reactive to the point that radical halogenation **using F causes an explosion** because of its high exothermicity. On the opposite side of the spectrum, **I does not react at all** because the entire reaction is net endothermic. However, **Cl and Br provide a good middle ground** between the two extremes, with Cl being relatively more reactive than Br.

Here is a PE diagram displaying a side-by-side comparison of each type of radical halogenation.



Core Concept 3.2: Hammond Postulate

As you can see, the distance of transition state 1 (TS1) from the starting point increases as step 1 becomes more endothermic. Here, we introduce the **Hammond Postulate**, a principle used to estimate the structure and energy of a TS. Because TSs exist for only a fraction of a second and cannot be directly observed, this postulate allows chemists to "infer" what they look like based on its energy level.

When a reaction is exothermic, the TS PE level stands closer to that of the reactants, thus the TS looks more like the reactants than it does the products (**early transition state**).

On the flipside, when a reaction is endothermic, the TS PE level stands closer to that of the products, thus the TS looks more like the products than it does the reactants (**late transition state**).

Here is a zoomed in snapshot of step 1 of the reactions for F and I.

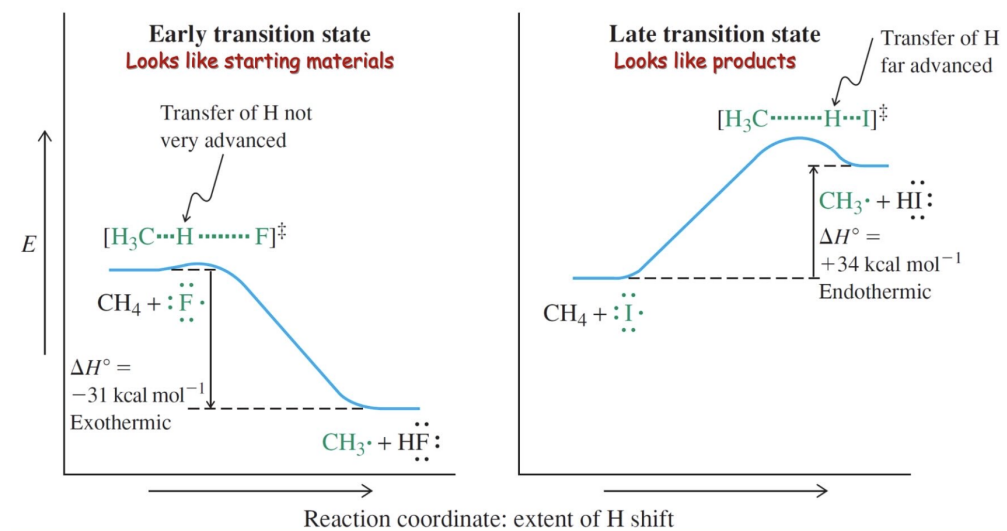


Figure 7: Zoomed Step 1 PE Diagrams for F and I

We can see that the TS1 for F is more reactant-like because the H that is being transferred from the methane to the F is closer to the methyl group than it is to the F. However, in the I reaction, the H is closer to the I than it is to the methyl group.

This can also be intuitively conceptualized by seeing how large the activation energies are for this reaction portion. Since the E_a is low for F, there would be less total energy inputted to separate the H from the methyl group, thus the TS would look like it hadn't shifted much but shifted enough to cross the threshold for the attractive force of F to overcome that of methyl and subsequently steal the H. In the case of the I reaction, since the E_a is high, there would be considerably more energy inputted to separate the H from the methyl group, thus the TS would look like the H had shifted a lot to cross the threshold for the attractive force of I (which is weaker than that of F) to overcome that of methyl and subsequently steal the H.

With this in mind, Br would also have a late transition state although not to the degree of I.

Core Concept 3.3: Regioselectivity

Using the Hammond Postulate, we can also explain the **regioselectivity** of the different halogenation reactions. Let's look at the different versions of isobutyl radical for example.



Figure 8: Isobutyl Radical Variations

Regioselectivity refers to the preferences of which carbon (just methyl, primary, secondary, or tertiary) the

halogen will replace the H on based on the stability of the semi radical transition state that the C-H develops into.

In the case of Cl halogenation, we see an **early TS where the radical is barely developed**, thus the stability of the primary isobutyl state (left) does not differ much from the tertiary isobutyl state (right). Thus, Cl halogenation is relatively indifferent to which C-H bond is being replaced.

As for Br halogenation, we see a **late TS where the radical is more developed**, thus the stability of the primary isobutyl state (left) is considerably less stable than that of the tertiary isobutyl state (right) due to the presence of more hyperconjugation. Thus, Br halogenation would favor the halogenation of the tertiary over the primary.

The ratio numbers we see in the lab for the different halogenation reactions across differently hyperconjugated carbons align with this explanation.

C-H Bond	F (25°C, gas)	Cl (25°C, gas)	Br(150°C, gas)
CH ₃ - H	0.5	0.004	0.002
RCH ₂ - H	1	1	1
R ₂ CH - H	1.2	4	80
R ₃ C - H	1.4	5	1700

Table 1: Your caption text here